VIT CHENNAI

Precision Agriculture Monitoring Quadruped Robot Dog

PROJECT I — PRE-FINAL REVIEW

A 12-DOF printed quadruped with a distributed field-sensing payload: an ESP32-S3 sensor node, an ESP32-CAM camera node and leaf-disease classification on the laptop, with locomotion validated in MuJoCo physics simulation.

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Problem Statement

- Field monitoring relies on manual scouting or drones; neither gives close-range, multi-parameter readings at crop level.
- Wheeled and tracked rovers struggle in mud, soft soil and the narrow furrows between planted rows.
- Disease is often noticed only once it is visible across the field, when treatment is costly.
- Commercial agricultural robots remain too expensive for most small and medium farms.

20–40 %

of global crop production is lost to plant pests and diseases every year (FAO).

Project Objectives

Mobility

A low-cost 3D-printed 12-DOF quadruped whose gait is validated in physics simulation against real servo torque limits.

Sensing

Close-range gas (TVOC, eCO_2), dust, temperature and distance sensing carried on the chassis.

Plant-health AI

Camera-based leaf-disease classification that first checks whether a leaf is in view at all.

Live dashboard

A wireless dashboard that logs every reading and tags each leaf image with the air conditions at capture time.

Design Revision Since Review III

The sensing payload moved off the Jetson onto two ESP32 nodes; locomotion moved to a physics simulator plus an MVP build.

Review III plan	Built now	Reason
Jetson Orin Nano on the robot	XIAO ESP32-S3 + ESP32-CAM nodes; inference on the laptop	A fall risks an expensive board; MG996R torque margin is thin; inference needs no onboard GPU
IMX219 CSI camera	ESP32-CAM (OV3660), Wi-Fi MJPEG stream	The ESP32-S3 has no MIPI-CSI camera interface
ROS 2 + MQTT on board	Flask dashboard over the laptop hotspot	Fewer moving parts; every device on one local network
Gazebo dynamics model	MuJoCo physics simulation	The Gazebo stance never stabilised; MuJoCo walks and reports metrics headless
Walking trials on hardware	Simulation results now; servo MVP next	Torque analysis showed walking needs CoM-centred footholds first

System Architecture

Sensor node

- XIAO ESP32-S3 + expansion board
- SGP30 gas · I²C 0x58
- MCP9808 temp · I²C 0x18
- PPD42 dust · D0
- HC-SR04 × 2 · D6–D9
- OLED, buzzer, button

Laptop · 192.168.137.1

- Wi-Fi hotspot, 2.4 GHz
- Flask dashboard :5000
- CSV log of every reading
- MobileNetV2 + leaf check
- Video relay to all viewers

Camera node

- ESP32-CAM, OV3660 sensor
- MJPEG video on :81
- Single photo on :80/capture
- Registers itself at boot

JSON every 5 s →

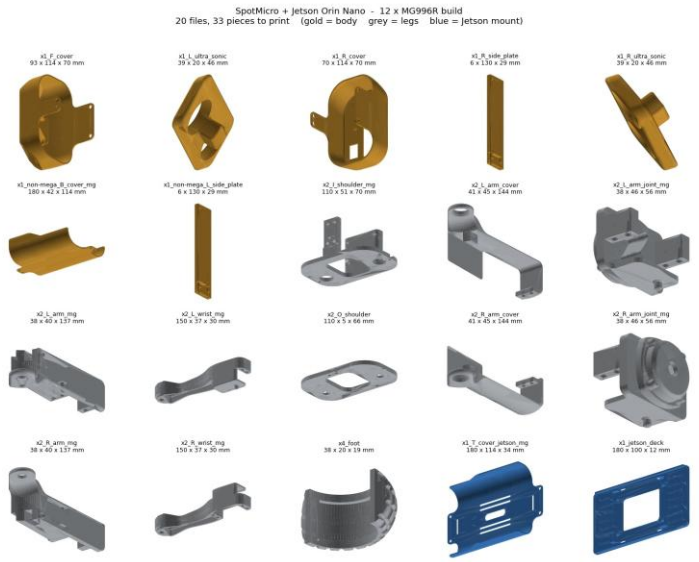
← MJPEG video, ~16–20 fps

Platform SpotMicro-based printed chassis carrying both nodes · 12 × MG996R servos (stand-and-step MVP) · gait and torque limits validated in MuJoCo

Any phone on the hotspot can open the dashboard at http://192.168.137.1:5000

Mechanical Build — Printed Chassis

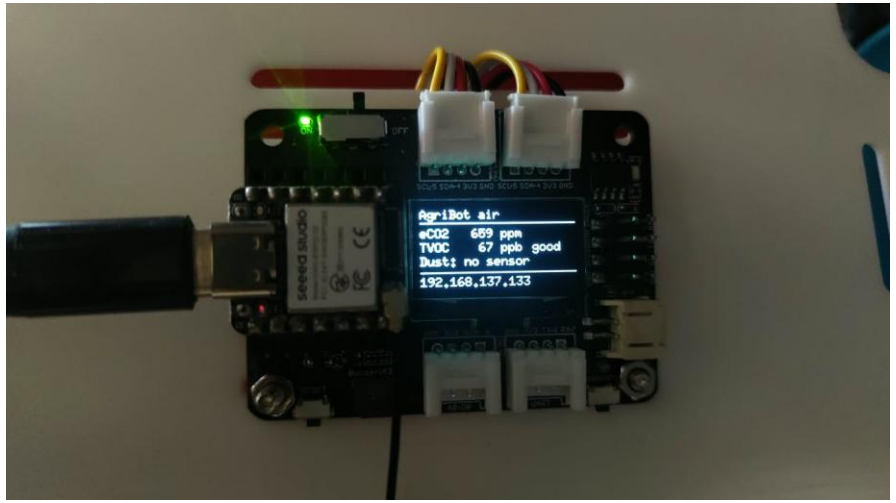
- 35 parts printed and verified; base design SpotMicro (KDY0523, CC-BY), adapted for our payload
- ~700 g PLA+ structure, TPU foot pads · printing cost ₹5,200
- Geometry measured from the prints: hip spacing 140 × 78 mm, femur 111 mm, tibia 125 mm
- Four team-designed parts: top cover, sensor deck, PPD42 and SGP30 brackets
- Sensor node now mounted on the printed top cover (photo)





Print set (left) · sensor node on the top cover (right)

Sensor Node Hardware



Live readings and IP address on the node's OLED

Every 5 V sensor output reaches the 3.3 V XIAO through a 1 kΩ / 2 kΩ divider: 5 V → 3.3 V, and the PPD42's 4 V → 2.7 V.

Sensor	Measures	Interface	Status
SGP30	TVOC, eCO_2	I^2C 0x58, Grove IIC 1	**Live**
MCP9808	Temperature ±0.25 °C	I^2C 0x18, Grove IIC 2	**Live**
OLED, buzzer	Local display, alerts	On board	**Live**
PPD42	Particles > 1 µm	D0 via divider	Wiring pending
HC-SR04 #1	Obstacle / row end	D6 trig, D7 echo	Wiring pending
HC-SR04 #2	Canopy height	D8 trig, D9 echo	Wiring pending

Firmware for the three pending sensors is written, compiled and uploaded; they report as absent until connected.

Firmware Engineering

Interrupt-timed dust

PPD42 low-pulse time is accumulated in an interrupt over 30 s windows, so sampling never blocks the loop.

Missing-sensor detection

An unplugged PPD42 read as 1,114,001 particles and sounded the alarm. A line held low with no pulses is now reported as absent.

Temperature-corrected range

Speed of sound from the MCP9808: $c = 331.3 + 0.606 \cdot T$ m/s. Fixed 343 m/s is ~3 % off at 35 °C.

Crosstalk-free ultrasonics

The two HC-SR04s fire alternately, never together, and each reading is median-of-3 filtered.

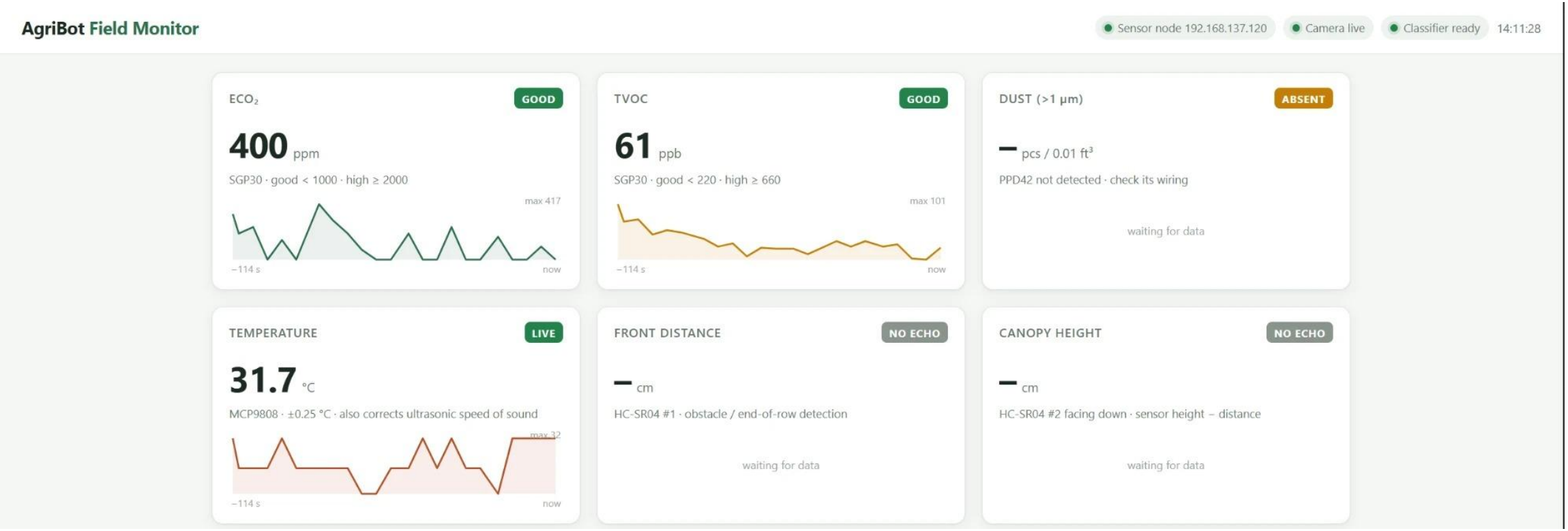
Wi-Fi diagnostics

Reports why a connection fails. It traced a −78 dBm link to a missing antenna; fitted, the link reads −26 dBm.

Self-registering camera

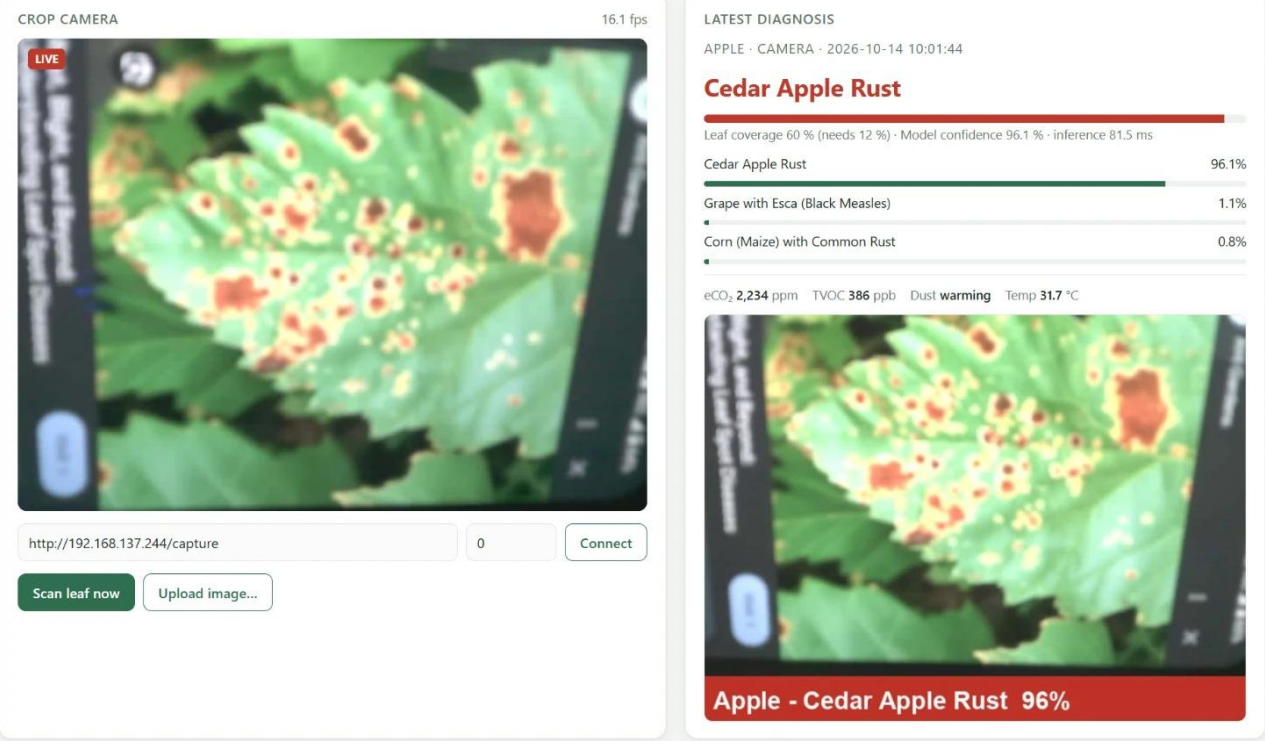
The ESP32-CAM sends its photo and stream URLs to the dashboard at boot, so a new IP address needs no setup.

Live Dashboard



Status bands TVOC per the German UBA guideline (220 / 660 ppb), eCO_2 1,000 / 2,000 ppm
Logging every reading to a daily CSV · 10-minute charts · no internet needed at runtime

Leaf-Disease Classification



Live scan through the ESP32-CAM: Cedar Apple Rust, 96 %, leaf coverage 60 %

1 Frame from the ESP32-CAM stream

2 Leaf check: ≥ 12 % of pixels leaf-coloured (hue 38–165°, saturation ≥ 0.2)

3 MobileNetV2, PlantVillage: 38 classes, 14 crops

4 Below 50 % confidence → reported as uncertain

5 Image saved, tagged with crop, condition, gas and temperature

Input	Result
Room, no leaf	Rejected (0–0.1 % coverage). Model alone: “Citrus Greening 82–90 %”
12 PlantVillage leaves	All accepted (27–76 % coverage)
Inference	30–80 ms per image on the laptop CPU

Results — Hardware Validation

Every working module was exercised on the bench; all values are from those runs.

Test	Result	What it shows
SGP30, breath test	eCO_2 400 → 746 ppm, TVOC 0 → 210 ppb within 5 s, back to baseline after	Gas sensing responds and recovers
MCP9808, finger contact	31.5 → 35.0 °C in ~10 s	Temperature tracks a real change
XIAO Wi-Fi link	−78 dBm with auth time-outs → −26 dBm with antenna fitted	Diagnostics found the fault
Camera video	~0.5 → 16–20 fps after moving to MJPEG streaming	Smooth live view for scanning
Camera onboarding	Registered itself on a new IP at every boot tested	No manual network setup
Leaf check	Room view rejected; 12 / 12 leaf images accepted	No disease reported without a leaf
Dust sensor absent	Reported as absent instead of a false 1.1 M count	No false alarms from wiring faults

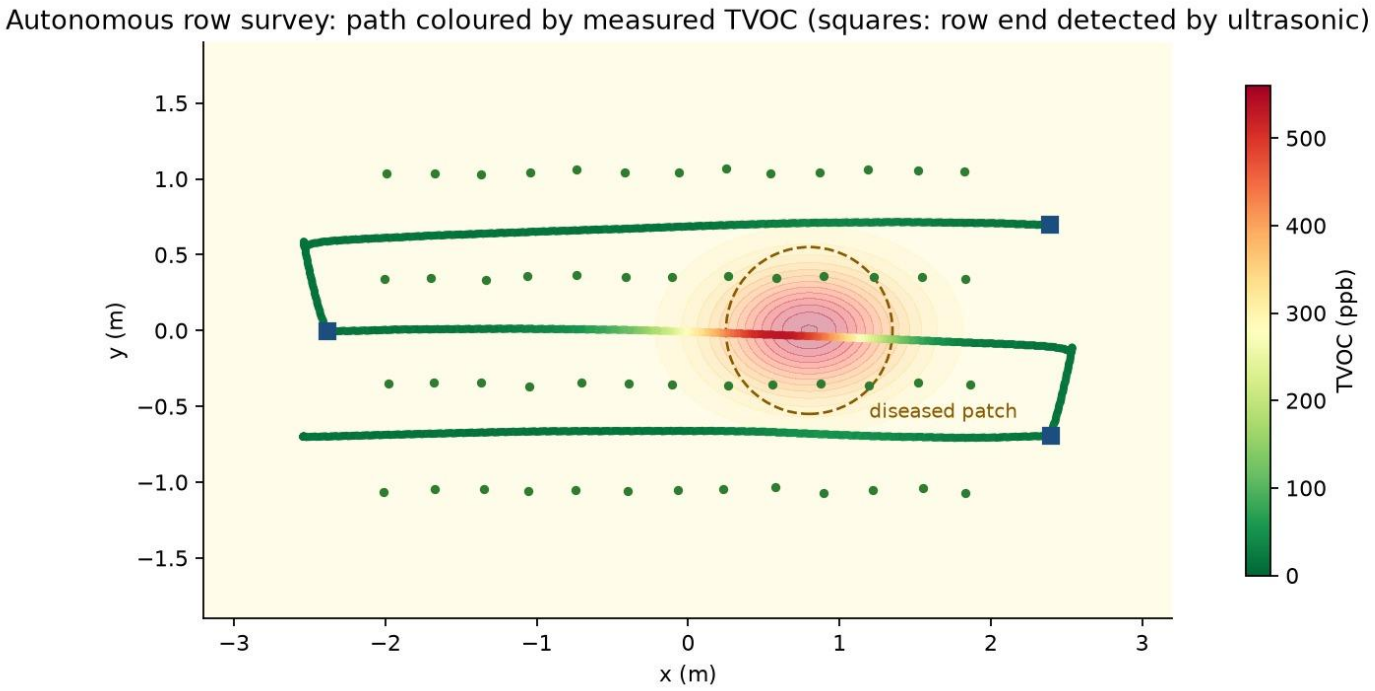
MuJoCo Physics Simulation



Surveying the middle row gap; the trail turns red over the diseased plants

- Robot from measured geometry: 140 × 78 mm hips, 111 / 125 mm legs, 1.35 kg
- 12 position servos capped at 1.0 N·m (MG996R stall 0.92–1.08 N·m)
- Leg IK checked against MuJoCo's own kinematics: 1,280 targets, 0.0000 mm error
- Mission: stand up, trot each row gap, detect row ends by ultrasonic, turn, log gas, dust and temperature

Results — Autonomous Field Survey



Path coloured by measured TVOC; squares mark row ends detected by the front ultrasonic

16.7 m · 129 s

all 3 row gaps surveyed, no falls

3 / 3

row ends detected by the front ultrasonic

≤ 2.7°

body tilt throughout; servos at their limit 1.3 % of the time

535 ppb

TVOC peak, 0.2 m from the true source (SGP30 lag)

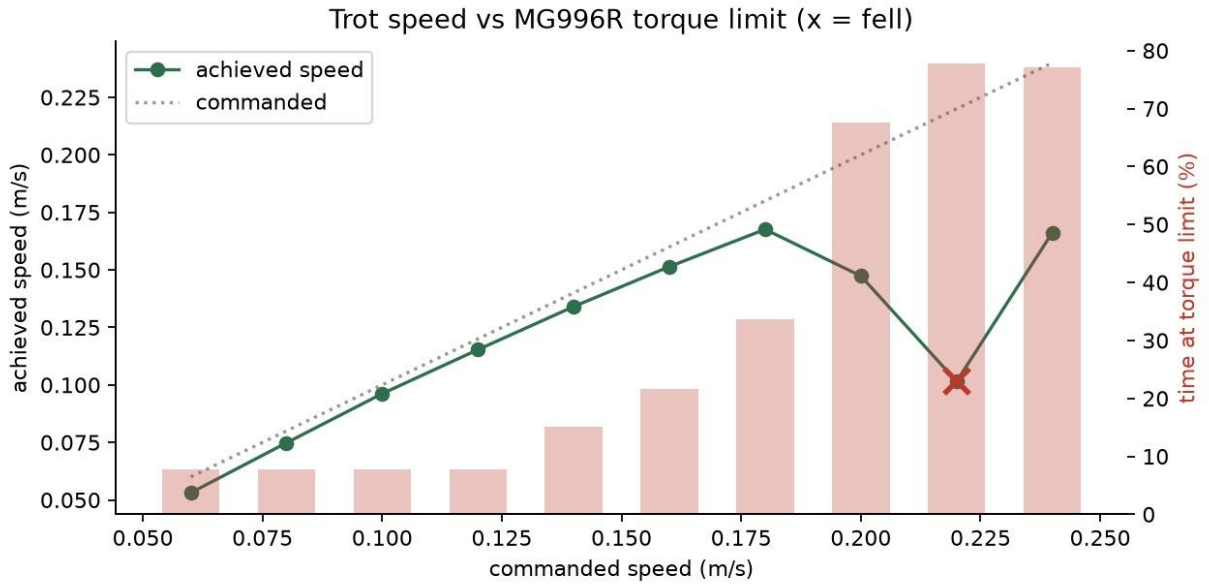
Engineering Findings

Centre of mass sits 12 mm aft

All four knees point backward, as on SpotMicro, so the leg mass hangs rearward.

With footholds centred on the body, the trot tipped backward within 1–3 s in all 24 gait settings tried.

Placing the feet around the measured centre of mass made it stable: 0.7° tilt trotting in place. The same correction applies to the physical robot.



Speed	Result
0.08–0.18 m/s	Tracks the command within 7 %
0.20 m/s	Servos at limit 68 % of the time, tilt 12°
0.22 m/s	**Falls**

Limitations

Limitation	Effect	How it is handled
Simulated gas, dust and temperature come from modelled fields	Simulation shows the method, not field data	Hardware sensors validated on the bench; field logging next
Classifier trained on lab photos (PlantVillage), 14 crops	Lower accuracy on real field images	Leaf check and 50 % confidence threshold; field-image fine-tuning as future work
SGP30 eCO_2 is estimated from VOCs; PPD42 counts particles, not PM2.5	Relative readings, not regulatory values	Used for trends and alerts, not absolute limits
Walking not yet shown on hardware; MG996R margins are thin	Locomotion evidence is simulation-only	Servo MVP before the final review, with the CoM correction ready
Dust and ultrasonic sensors not yet wired	Not part of today's live demo	Firmware complete and uploaded; wiring next

Remaining Work Before the Final Review

01 Wire the PPD42 and both HC-SR04s through their dividers; firmware already handles them

02 Grove Relay v1.3: act on readings (irrigation pump or gas alert), with a dashboard control

03 Servo MVP: stand, sit and step, using the simulator's gait and centre-of-mass correction

Deliverables

Project report in the BCSE497J template, with similarity ≤ 15 %, by 21 October

Journal / conference paper submission with acknowledgement

Final panel review demonstration

Technology Stack

Hardware

Seeed XIAO ESP32-S3 + expansion board · ESP32-CAM (OV3660) + MB · SGP30, MCP9808, PPD42, HC-SR04 × 2 · 12 × MG996R

Firmware

Arduino-ESP32 3.3 · Adafruit SGP30 · U8g2 · esp_http_server (MJPEG)

Dashboard & AI

Python · Flask · PyTorch · Hugging Face Transformers · MobileNetV2 (PlantVillage)

Simulation

MuJoCo 3 · NumPy · Matplotlib · headless video and metrics

Mechanical

SpotMicro base (CC-BY) · PLA+ / TPU · parametric Python generators for the custom parts

Collaboration

Git + GitHub: firmware, dashboard, simulation, print files and results in one repository

Literature Review (18 Papers)

#	Authors & Year	Paper Title	Venue	Key Contribution	Research Gap
1	Bledt et al. (2018)	MIT Cheetah 3: Design and Control of a Robust, Dynamic Quadruped Robot	IEEE/RSJ IROS	Designed high-torque density proprioceptive actuators and Model Predictive Control (MPC) for dynamic walking over rough terrain without vision.	High cost (~$50k+), complex custom planetary gearing, no environmental/agricultural sensing payload.
2	Kau et al. (2019)	Stanford Doggo: An Open-Source, Quasi-Direct-Drive Quadruped	IEEE ICRA	Developed an open-source 3D-printed quadruped platform using quasi-direct drive brushless motors for dynamic bounding.	Optimized purely for acrobatic jumps/speed; lacks on-board compute for perception and has <10 min battery life.
3	Katz et al. (2019)	Mini Cheetah: A Hardware Design for Extremity Locomotion	IEEE ICRA	Modular 3-DOF leg design using low-cost backdrivable BLDC motors enabling agile omnidirectional gaits.	Focuses strictly on mechanical agility in laboratory environments; no autonomous navigation or edge AI pipeline.
4	Hutter et al. (2016)	ANYmal - A Unique Quadruped Robot for Autonomous Operation in Harsh Environments	IEEE/RSJ IROS	Industrial-grade quadruped with series elastic actuators (SEAs) and IP67 ruggedized body for outdoor inspection.	Prohibitively expensive ($100k+), heavy industrial form-factor unsuitable for smallholder precision farming.
5	Ding et al. (2021)	Learning to Walk on Uneven Terrain for Quadrupedal Robots using Reinforcement Learning	IEEE Transactions on Robotics (T-RO)	Trained neural network locomotion policies in simulation (Isaac Gym) with robust sim-to-real transfer over rugged slopes.	High computational training overhead; policy does not integrate multi-sensor agronomic payloads or path tracking.

Literature Review (18 Papers)

#	Authors & Year	Paper Title	Venue	Key Contribution	Research Gap
6	Bellicoso et al. (2018)	Dynamic Locomotion Through Online Nonlinear Model Predictive Control for Quadrupedal Robots	IEEE Robotics & Automation Letters (RA-L)	Real-time ZMP and CoM trajectory optimization enabling adaptive crawl and trot gaits over unmodeled obstacles.	Requires high-end x86 computer boards; difficult to run concurrently with edge vision models on low-power platforms.
7	Fehlmann et al. (2023)	Precision Agriculture Using Autonomous Ground Robots: A Review	Springer Precision Agriculture	Systematic review of wheeled and tracked agricultural rovers for weed control, phenotyping, and soil sampling.	Identifies severe mobility failure of wheeled rovers in wet mud, soft soil, and high-crop canopy furrows.
8	Roldán et al. (2018)	Mini-UAVs and Mobile Robots for Precision Agriculture: A Review of Multi-Robot Systems	Computers and Electronics in Agriculture (Elsevier)	Analyzed complementary roles of aerial drones (macro-surveys) and ground robots (micro-inspection).	Highlighted drone limitations: 20-min flight time, inability to inspect leaf undersides, and blind spots beneath crop canopies.
9	Bogue, R. (2016)	Robots in Agriculture: State of the Art and Future Directions	Industrial Robot: An International Journal	Surveyed commercial adoption of agricultural robotics, highlighting high ROI in crop health monitoring.	Commercial units are bulky, heavy, cause soil compaction, and require flat structured greenhouse tracks.
10	Vasconez et al. (2020)	Human–Robot Interaction in Agriculture: A Survey and Review	Computers and Electronics in Agriculture (Elsevier)	Investigated telemetry, UX dashboards, and safety mechanisms for farmer-robot interfaces in open fields.	Found most agricultural robotic interfaces are academic prototypes lacking real-time edge telemetry and alert maps.

Literature Review (18 Papers)

#	Authors & Year	Paper Title	Venue	Key Contribution	Research Gap
11	Mohanty et al. (2016)	Using Deep Learning for Image-Based Plant Disease Detection	Frontiers in Plant Science	Trained AlexNet and GoogLeNet on 54,306 images across 26 diseases with 99.35% classification accuracy.	Tested only on static laboratory-controlled leaf images with uniform backgrounds; lacks edge device deployment.
12	Ferentinos, K. P. (2018)	Deep Learning Models for Plant Disease Detection and Diagnosis	Computers and Electronics in Agriculture (Elsevier)	Evaluated VGG, ResNet, and AlexNet on 87,848 leaf images with real-world noise variations (99.5% accuracy).	High parameter count (VGG/ResNet) causes severe latency and memory bottlenecks on embedded single-board computers.
13	Too et al. (2019)	A Comparative Study of Fine-Tuning Deep Learning Architectures for Plant Disease Identification	Computers and Electronics in Agriculture (Elsevier)	Benchmarked MobileNetV2, DenseNet-121, and InceptionV3; proved MobileNetV2 achieves 97.4% accuracy with 10× fewer FLOPs.	Did not perform hardware-specific quantization (TensorRT / ONNX) or test real-time video stream ingestion.
14	Chen et al. (2020)	Using Deep Transfer Learning for Image-Based Plant Disease Identification on Embedded Devices	Computers and Electronics in Agriculture (Elsevier)	Deployed lightweight CNN on Raspberry Pi 3B+ using INT8 quantization, achieving 12 FPS at 92.1% accuracy.	Insufficient FPS for continuous mobile robot navigation; no integration with robotic gait or sensor fusion.
15	Saleem et al. (2019)	Plant Disease Detection and Classification by Deep Learning: A Review	Computers and Electronics in Agriculture (Elsevier)	Comprehensive meta-analysis of deep learning architectures, dataset biases, and real-time field challenges.	Concluded that field-ready solutions require mobile ground robots that bring cameras within 0.5m of plant leaves.

Literature Review (18 Papers)

#	Authors & Year	Paper Title	Venue	Key Contribution	Research Gap
16	Ray, P. P. (2017)	Internet of Things for Smart Agriculture: Technologies, Practices and Future Direction	Journal of Ambient Intelligence and Smart Environments	Proposed layered IoT architecture for soil moisture, microclimate, and particulate monitoring in precision farming.	Relied exclusively on stationary sensor nodes (motes), creating spatial coverage gaps across large farm tracts.
17	Elijah et al. (2018)	An Overview of Internet of Things (IoT) and Data Analytics in Smart Agriculture	IEEE Internet of Things Journal	Investigated wireless sensor networks (WSNs) and cloud data pipelines for environmental anomaly detection.	High infrastructure cost of deploying hundreds of stationary nodes; vulnerable to field vandalism and weather damage.
18	Bacco et al. (2019)	Smart Farming: Technologies, Applications and Challenges in Precision Agriculture	Computers and Electronics in Agriculture (Elsevier)	Explored multi-sensor data fusion (temperature, humidity, organic gases, optical imagery) for soil health profiling.	Lacked mobile autonomous platforms to transport multi-sensor payloads dynamically throughout crop rows.

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[2] Kau, N., et al. (2019). Stanford Doggo: An Open-Source, Quasi-Direct-Drive Quadruped. IEEE ICRA. DOI: 10.1109/ICRA.2019.8794436

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[11] Mohanty, S. P., et al. (2016). Using Deep Learning for Image-Based Plant Disease Detection. Frontiers in Plant Science, 7, 1419. DOI: 10.3389/fpls.2016.01419

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[14] Chen, J., et al. (2020). Using Deep Transfer Learning for Image-Based Plant Disease Identification on Embedded Devices. Computers and Electronics in Agriculture, 176, 105653.

[15] Saleem, M. H., et al. (2019). Plant Disease Detection and Classification by Deep Learning: A Review. Computers and Electronics in Agriculture, 164, 104902.

[16] Ray, P. P. (2017). Internet of Things for Smart Agriculture: Technologies, Practices and Future Direction. Journal of Ambient Intelligence and Smart Environments, 9(4), 395–420.

[17] Elijah, O., et al. (2018). An Overview of Internet of Things (IoT) and Data Analytics in Smart Agriculture. IEEE Internet of Things Journal, 5(5), 3758–3773.

[18] Bacco, M., et al. (2019). Smart Farming: Technologies, Applications and Challenges. Computers and Electronics in Agriculture, 164, 104908.

Conclusion

- The sensing payload works end to end: two wireless nodes, a live dashboard, logging, and leaf-disease classification that refuses to guess without a leaf.
- Locomotion is validated in physics simulation with real servo limits: a full autonomous row survey without a fall.
- Simulation produced two findings for the physical build: centre-of-mass-centred footholds, and a ~0.18 m/s speed ceiling for MG996R servos.

Repository github.com/AdithyaSM31/Precision-Agriculture-Monitoring-Quadruped-Robot-Dog

Thank you